What Is a Motor?

A motor is an electrical machine that converts electrical energy into mechanical energy.

In simple terms, a motor uses electricity to turn a shaft. That rotation can drive a machine, fan, wheel, drone, or cable drum.

1. How Does a Motor Work?

A motor works mainly through the interaction between magnetic fields and electric current.

Here's the process in five steps:

1. An electrical supply is connected to the motor.
2. Current flows through the motor windings.
3. The windings produce a magnetic field.
4. The interaction between magnetic fields creates torque on the rotor.
5. The rotor turns, causing the attached shaft to rotate.

For example, when you connect a DC motor to a 12 V battery, its shaft begins to turn.

2. What Are the Main Types of Motors?

Motors can be classified in different ways. By the type of electrical supply, there are two main categories:

Electric Motor
- AC Motor (Alternating Current)
  - Induction Motor
  - Synchronous Motor
- DC Motor (Direct Current)
  - Brushed DC Motor
  - Brushless DC Motor (BLDC)
  - Servo Motor
  - Stepper Motor

This is a simplified classification. A servo motor can also refer to a complete motor-control system.

3. Motor Types, How They Work, Uses, Advantages, and Disadvantages

A. AC Motor (Alternating Current Motor)

How it works:

- When AC power is supplied, the stator produces a rotating magnetic field.
- This magnetic field induces current in the rotor.
- The interaction between the magnetic fields makes the rotor turn.

Uses:

- Industrial machinery
- Water pumps
- Fans
- Compressors
- Conveyors

Advantages:

- Strong and reliable
- Low maintenance
- Well suited to industrial applications

Disadvantages:

- A variable-frequency drive (VFD) is needed for speed control in many applications.
- It cannot run directly from a battery.
- Some types draw high current during startup.

B. Brushed DC Motor

How it works:

- DC power is supplied by a battery.
- Current flows through carbon brushes and a commutator into the armature windings.
- The interaction of the magnetic fields makes the rotor turn.

Uses:

- Small robots
- Electric toys
- Small pumps
- DC gear motors
- Cable drum mechanisms

Advantages:

- Simple construction
- Speed is easy to control
- Low cost
- Can provide good starting torque

Disadvantages:

- The brushes wear out.
- The motor needs maintenance.
- It can produce sparks and heat.

C. BLDC Motor (Brushless DC Motor)

How it works:

- It has no mechanical brushes.
- The stator contains multiple windings.
- An electronic controller, such as an ESC, energizes the windings in the correct sequence.
- The resulting magnetic field turns the rotor.

Uses:

- Drones
- Electric vehicles
- Cooling fans
- Robotics
- Cable drum drive systems

Advantages:

- High efficiency
- High power-to-weight ratio
- Low maintenance
- High-speed capability
- Long service life

Disadvantages:

- Requires an ESC or other electronic controller.
- Can cost more than a brushed motor.
- Its controller and wiring are more complex.

D. Servo Motor

A servo motor includes a motor and a feedback sensor, such as an encoder.

How it works:

1. The controller sends a desired position or speed command.
2. The encoder reports the actual position.
3. The controller compares the desired and actual positions.
4. The controller adjusts the motor based on the difference.

Uses:

- Robotic arms
- CNC machines
- Aircraft control surfaces
- Robotic joints

Advantages:

- Accurate position control
- Accurate speed control
- Closed-loop feedback

Disadvantages:

- Can be expensive
- The control system is more complex
- Setup and tuning are required

E. Stepper Motor

A stepper motor turns in small, fixed angular steps instead of rotating continuously.

For example, one command might turn the motor by 1.8°. The actual step angle depends on the motor’s specifications.

Uses:

- 3D printers

- CNC machines
- Camera positioning
- Robotics

Advantages:

- Precise step-by-step movement
- Simple positioning control
- Good holding torque at low speeds

Disadvantages:

- Torque decreases at high speeds.
- The motor can miss steps.
- It may generate substantial heat in some operating conditions.